MORPHISMS OF LOCALLY RINGED SMALL ETALE TOPOI OF AFFINE SCHEMES COME FROM MORPHISMS OF SCHEMES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for existence and uniqueness: the affine computations, the stalk check, the functoriality, the identification of pushforwards and the comparison of structural maps are all inside the counted span.

Editorial note. Every labelled reference occurring inside the counted spans points to a chapter of the Project that is not part of this card's source tree. Each is carried in assembly.external_references as a non-load-bearing navigation pointer whose source location is the invocation inside the counted span, so no load-bearing mathematics is replaced by a stub.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Small etale topos of a scheme with its structure sheaf and the associated locally ringed topos. Same chapter, the topos Sh(X_etale) together with the structure sheaf, and the morphism (f_{small}, f_{small}^sharp) attached to a morphism of schemes. Supplies the objects compared by the asserted 2-isomorphism and the target of the construction.
- Morphism of locally ringed topoi and 2-isomorphism of ringed topoi. Modules on Sites, the definition of a morphism of ringed topoi, of a 2-morphism between them, and of the locally ringed condition. Supplies the language in which the statement is phrased and the compatibility that the last paragraph of the proof checks.
- Sections of the structure sheaf as maps h_V -> O_X and the Cech-style computation of maps out of representable sheaves. Same chapter, the description of the sheaf h_V represented by an object of the small etale site and of morphisms from it, together with the fact that the small etale site has enough points. Supplies the computation of the isomorphism g^{-1}h_V = h_{X x_Y V} and its functoriality, which produce the morphism of schemes f.

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- Uniqueness of the morphism of schemes underlying a morphism of topoi. Same chapter, the result that the small morphism of ringed topoi determines the morphism of schemes. Supplies the uniqueness clause of the statement.

Source reference. Original lemma 04IC: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 5361–5361. Navigation pointer invoked at proof line 5361; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00U9: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 5382–5382. Navigation pointer invoked at proof line 5382; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original theorem 03PU: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 3546–3546. Navigation pointer invoked at proof line 5428; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 04HU: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 3610–3610. Navigation pointer invoked at proof line 5433; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 04GX: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 5448–5448. Navigation pointer invoked at proof line 5448; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 04HR: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 5290–5290. Navigation pointer invoked at proof line 5290; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let X, Y be affine schemes. Let*

$$(g, g^\#) : (Sh(X_{étale}), \mathcal{O}_X) \longrightarrow (Sh(Y_{étale}), \mathcal{O}_Y)$$

be a morphism of locally ringed topoi. Then there exists a unique morphism of schemes $f : X \to Y$ such that $(g, g^\#)$ is 2-isomorphic to $(f_{small}, f^\sharp_{small})$, see Modules on Sites, Definition 04IC.

Proof. In this proof we write $\mathcal{O}_X$ for the structure sheaf of the small étale site $X_{étale}$, and similarly for $\mathcal{O}_Y$. Say $Y = \mathrm{Spec}(B)$ and $X = \mathrm{Spec}(A)$. Since $B = \Gamma(Y_{étale}, \mathcal{O}_Y)$, $A = \Gamma(X_{étale}, \mathcal{O}_X)$ we see that $g^\sharp$ induces a ring map $\varphi : B \to A$. Let $f = \mathrm{Spec}(\varphi) : X \to Y$ be the corresponding morphism of affine schemes. We will show this f does the job.

Let $V \to Y$ be an affine scheme étale over Y. Thus we may write $V = \mathrm{Spec}(C)$ with C an étale B-algebra. We can write

$$C = B[x_1, \ldots, x_n]/(P_1, \ldots, P_n)$$

with P_i polynomials such that $\Delta = \det(\partial P_i/\partial x_j)$ is invertible in C, see for example Algebra, Lemma 00U9. If T is a scheme over Y, then a T-valued point of V is given by n sections of $\Gamma(T, \mathcal{O}_T)$ which satisfy the polynomial equations $P_1 = 0, \ldots, P_n = 0$. In other words, the sheaf h_V on $Y_{étale}$ is the equalizer of the two maps

$$\prod_{i=1,\ldots,n} \mathcal{O}_Y \xrightarrow[b]{a} \prod_{j=1,\ldots,n} \mathcal{O}_Y$$

where $b(h_1, \ldots, h_n) = 0$ and $a(h_1, \ldots, h_n) = (P_1(h_1, \ldots, h_n), \ldots, P_n(h_1, \ldots, h_n))$. Since g^{-1} is exact we conclude that the top row of the following solid commutative diagram is an equalizer diagram as well:

$$\begin{array}{ccccc} g^{-1}h_V & \longrightarrow & \prod_{i=1,\ldots,n} g^{-1}\mathcal{O}_Y & \xrightarrow[g^{-1}b]{g^{-1}a} & \prod_{j=1,\ldots,n} g^{-1}\mathcal{O}_Y \\ \vdots & & \downarrow \prod g^\sharp & & \downarrow \prod g^\sharp \\ h_{X\times_Y V} & \longrightarrow & \prod_{i=1,\ldots,n} \mathcal{O}_X & \xrightarrow[b']{a'} & \prod_{j=1,\ldots,n} \mathcal{O}_X \end{array}$$

Here b' is the zero map and a' is the map defined by the images $P_i' = \varphi(P_i) \in A[x_1, \ldots, x_n]$ via the same rule $a'(h_1, \ldots, h_n) = (P_1'(h_1, \ldots, h_n), \ldots, P_n'(h_1, \ldots, h_n))$. that a was defined by. The commutativity of the diagram follows from the fact that $\varphi = g^\sharp$ on global sections. The lower row is an equalizer diagram also, by exactly the same arguments as before since $X \times_Y V$ is the affine scheme $\mathrm{Spec}(A \otimes_B C)$ and $A \otimes_B C = A[x_1, \ldots, x_n]/(P_1', \ldots, P_n')$. Thus we obtain a unique dotted arrow $g^{-1}h_V \to h_{X\times_Y V}$ fitting into the diagram

We claim that the map of sheaves $g^{-1}h_V \to h_{X\times_Y V}$ is an isomorphism. Since the small étale site of X has enough points (Theorem 03PU) it suffices to prove this on stalks. Hence let $\overline{x}$ be a geometric point of X, and denote p the associate point of the small étale topos of X. Set $q = g \circ p$. This is a point of the small étale topos of Y. By Lemma 04HU we see that q corresponds to a geometric point $\overline{y}$ of Y. Consider the map of stalks

$$(g^\sharp)_p : (\mathcal{O}_Y)_{\overline{y}} = \mathcal{O}_{Y,q} = (g^{-1}\mathcal{O}_Y)_p \longrightarrow \mathcal{O}_{X,p} = (\mathcal{O}_X)_{\overline{x}}$$

Since $(g, g^\sharp)$ is a morphism of *locally* ringed topoi $(g^\sharp)_p$ is a local ring homomorphism of strictly henselian local rings. Applying localization to the big commutative diagram above and Algebra, Lemma 04GX we conclude that $(g^{-1}h_V)_p \to (h_{X\times_Y V})_p$ is an isomorphism as desired.

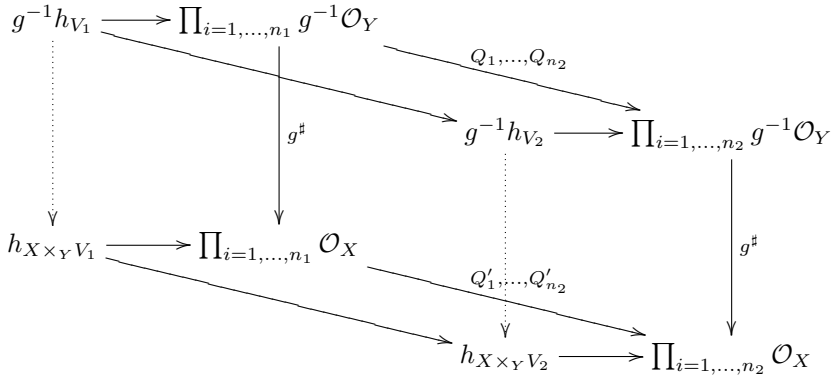
We claim that the isomorphisms $g^{-1}h_V \to h_{X\times_Y V}$ are functorial. Namely, suppose that $V_1 \to V_2$ is a morphism of affine schemes étale over Y. Write $V_i = \mathrm{Spec}(C_i)$ with

$$C_i = B[x_{i,1}, \ldots, x_{i,n_i}]/(P_{i,1}, \ldots, P_{i,n_i})$$

The morphism $V_1 \to V_2$ is given by a B-algebra map $C_2 \to C_1$ which in turn is given by some polynomials $Q_j \in B[x_{1,1}, \ldots, x_{1,n_1}]$ for $j = 1, \ldots, n_2$. Then it is an easy matter to show that the diagram of sheaves

$$\begin{array}{ccc} h_{V_1} & \longrightarrow & \prod_{i=1,\ldots,n_1} \mathcal{O}_Y \\ \downarrow & & \downarrow Q_1,\ldots,Q_{n_2} \\ h_{V_2} & \longrightarrow & \prod_{i=1,\ldots,n_2} \mathcal{O}_Y \end{array}$$

is commutative, and pulling back to $X_{étale}$ we obtain the solid commutative diagram



where $Q'_j \in A[x_{1,1}, \ldots, x_{1,n_1}]$ is the image of Q_j via φ. Since the dotted arrows exist, make the two squares commute, and the horizontal arrows are injective we see that the whole diagram commutes. This proves functoriality (and also that the construction of $g^{-1}h_V \to h_{X\times_Y V}$ is independent of the choice of the presentation, although we strictly speaking do not need to show this).

At this point we are able to show that $f_{small,*} \cong g_*$. Namely, let $\mathcal{F}$ be a sheaf on $X_{étale}$. For every $V \in \mathrm{Ob}(X_{étale})$ affine we have

$$\begin{aligned}
(g_*\mathcal{F})(V) &= \mathrm{Mor}_{Sh(Y_{étale})}(h_V, g_*\mathcal{F}) \\
&= \mathrm{Mor}_{Sh(X_{étale})}(g^{-1}h_V, \mathcal{F}) \\
&= \mathrm{Mor}_{Sh(X_{étale})}(h_{X\times_Y V}, \mathcal{F}) \\
&= \mathcal{F}(X \times_Y V) \\
&= f_{small,*}\mathcal{F}(V)
\end{aligned}$$

where in the third equality we use the isomorphism $g^{-1}h_V \cong h_{X\times_Y V}$ constructed above. These isomorphisms are clearly functorial in $\mathcal{F}$ and functorial in V as the isomorphisms $g^{-1}h_V \cong h_{X\times_Y V}$ are functorial. Now any sheaf on $Y_{étale}$ is determined by the restriction to the subcategory of affine schemes (Topologies, Lemma 04HR), and hence we obtain an isomorphism of functors $f_{small,*} \cong g_*$ as desired.

Finally, we have to check that, via the isomorphism $f_{small,*} \cong g_*$ above, the maps $f^\sharp_{small}$ and $g^\sharp$ agree. By construction this is already the case for the global sections of $\mathcal{O}_Y$, i.e., for the elements of B. We only need to check the result on sections over an affine V étale over Y (by Topologies, Lemma 04HR again). Writing $V = \mathrm{Spec}(C)$, $C = B[x_i]/(P_j)$ as before it suffices to check that the coordinate functions x_i are mapped to the same sections of $\mathcal{O}_X$ over $X \times_Y V$. And this is exactly what it means that the diagram

$$\begin{array}{ccc}
g^{-1}h_V & \longrightarrow & \prod_{i=1,\ldots,n} g^{-1}\mathcal{O}_Y \\
\downarrow & & \downarrow \prod g^\sharp \\
h_{X\times_Y V} & \longrightarrow & \prod_{i=1,\ldots,n} \mathcal{O}_X
\end{array}$$

commutes. Thus the lemma is proved. □